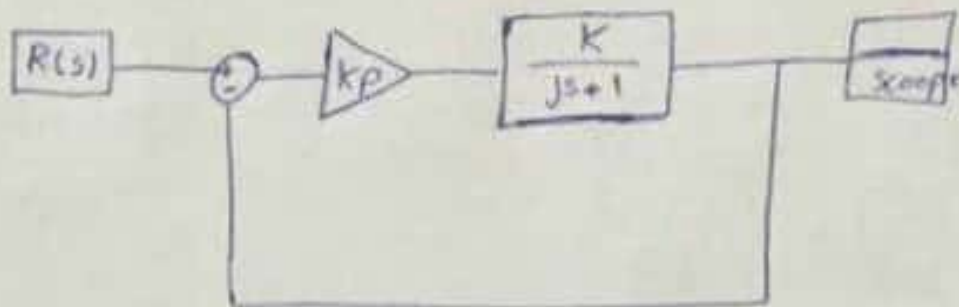


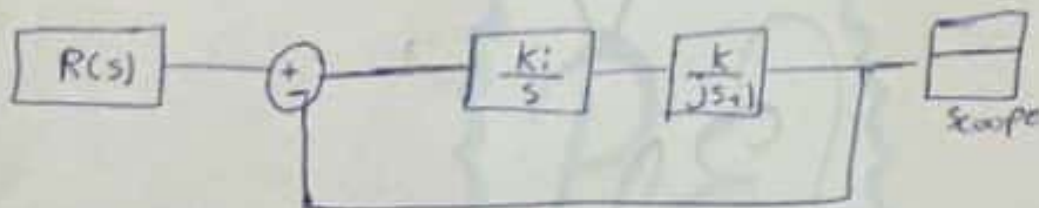
- second order system

A) Varying k_p in P controller



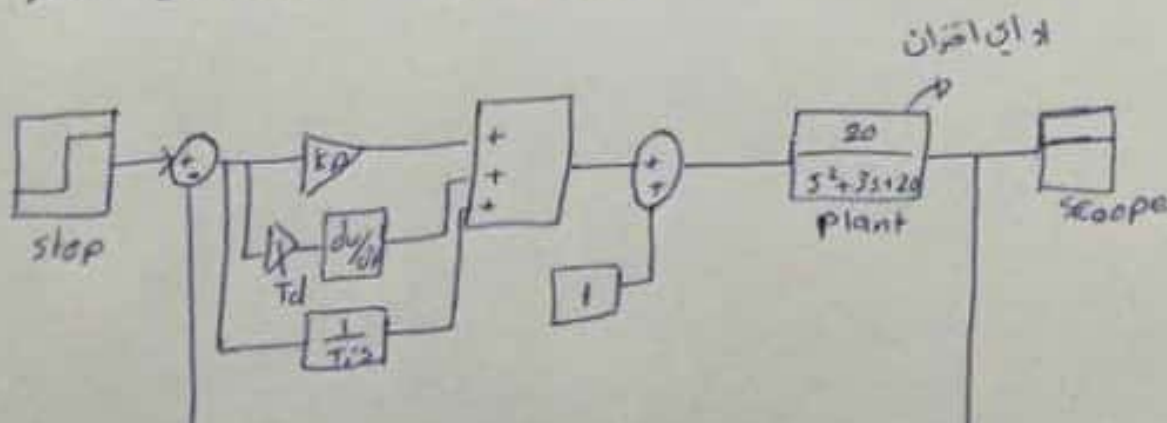
when increase $k_p \uparrow$ the overshoot increase
and Reduce steady state error

B) Varying k_i in I controller

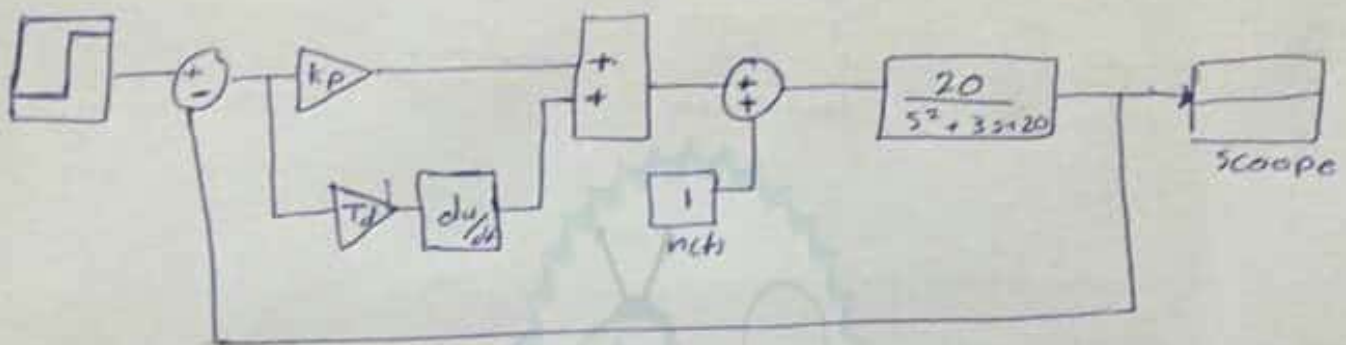


when k_i increase
the steady state error = zero

C) Varying T_i in PID controller if $k_p=5$ and $T_d=10$ to reach 20% of maximum overshoot



D) varying T_d in PD controller if $k_p=10$



When T_d increase in PD will reduce the oscillation and reduce the steady state error

Given $G(s) = \frac{1}{s(s+0.11)}$, negative unity feedback

unit step input, a) simulate this system

b) plot the output of the system

c) discuss the result ; d) select a controller
verify your selection

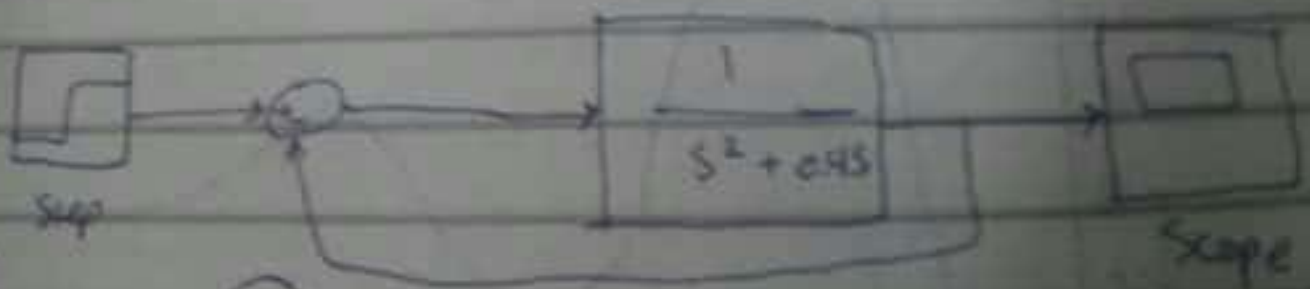
e) simulate the controlled system

f) plot and explain the response of controlled system

P.2 plot and explain the full load armature current

of a servo system.

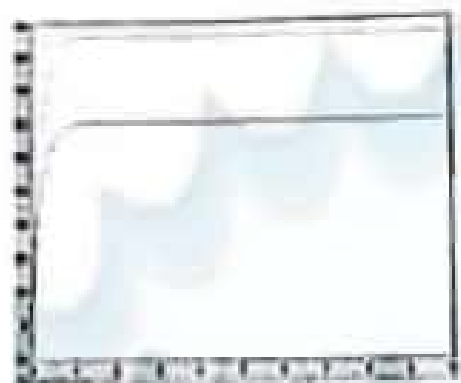
Solution:-



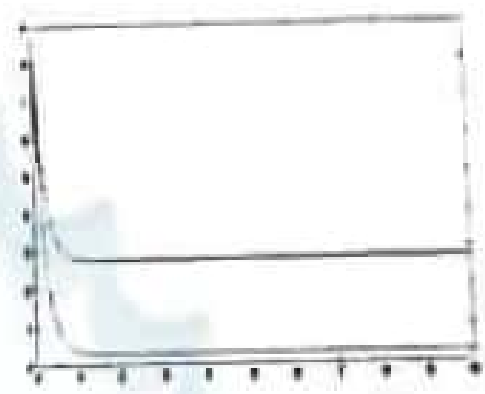
Q1: [10 marks] For second order system plot the simulation and output and steady output.

Circuit	output	Command and type of response
Varying damping ratio [0.5 / 1 / 4]		
Varying damping ratio [0 / -0.1]		
Varying Kp [2 . 6]		
If Kp=10 Varying Td [1 . 6]		

Q2) (10 marks) The following curves are the output of dc motor (open loop -)



Speed vs Time



Current vs Time

- Which plot represents the case of no load in each curve
- Explain the effect of load on i_a and w
- Sketch i_a and w in case of position closed loop system on the above case

Q3) (10 marks) a) Determine the state space matrices A,B,C & D for a system given

$$Y'''(t) + 4Y''(t) + 0.5Y' = Zr(t)$$

Parameters

A:

B:

C:

D:

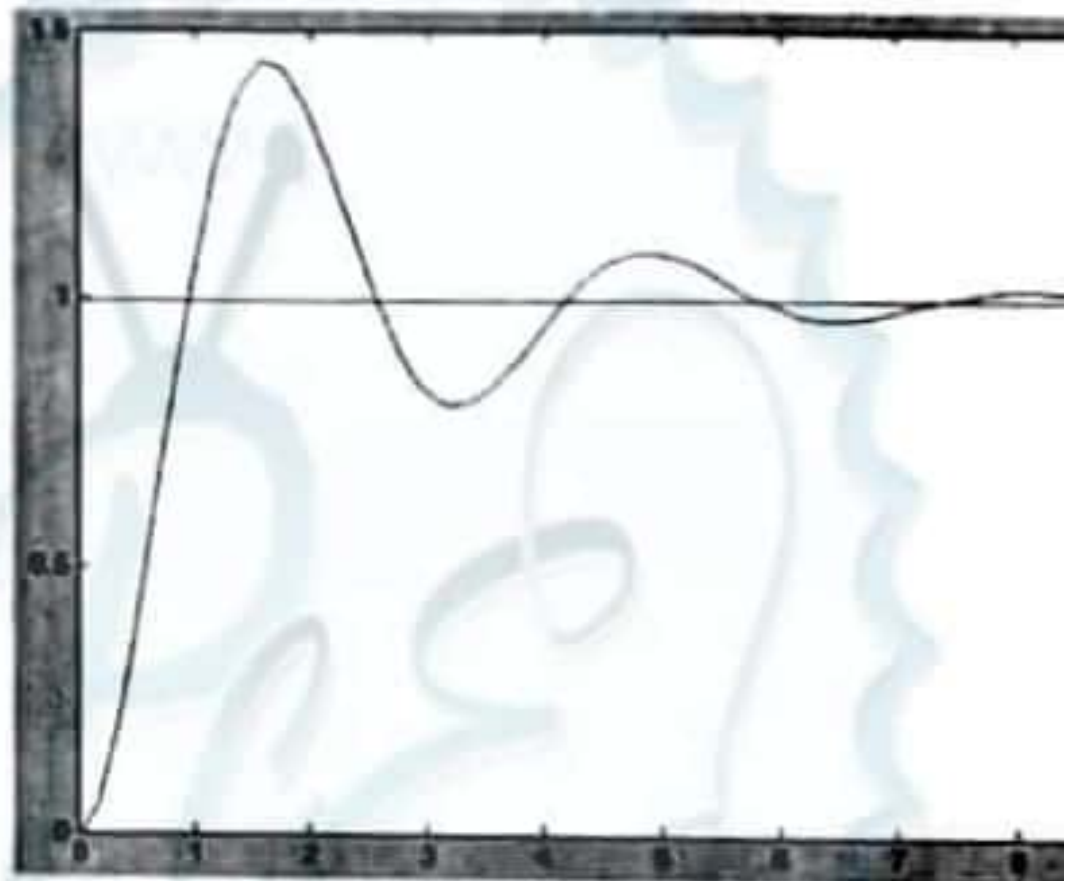
b) Find the transfer function of above equation

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Name: _____

Q2) [10 marks] From the plotted curve obtain the time response specifications of the curve

td	tr	tp	Mp



Q3) [10 marks]

Explain

A. Increase proportional controller in first order system
when increase K_P

B. Increase integral controller in second order system

C. Increase Time constant in first order system

Q2) [10 marks] The unit step response of a second order system is shown for $\omega_n = 1$ and $\zeta = 0.5$

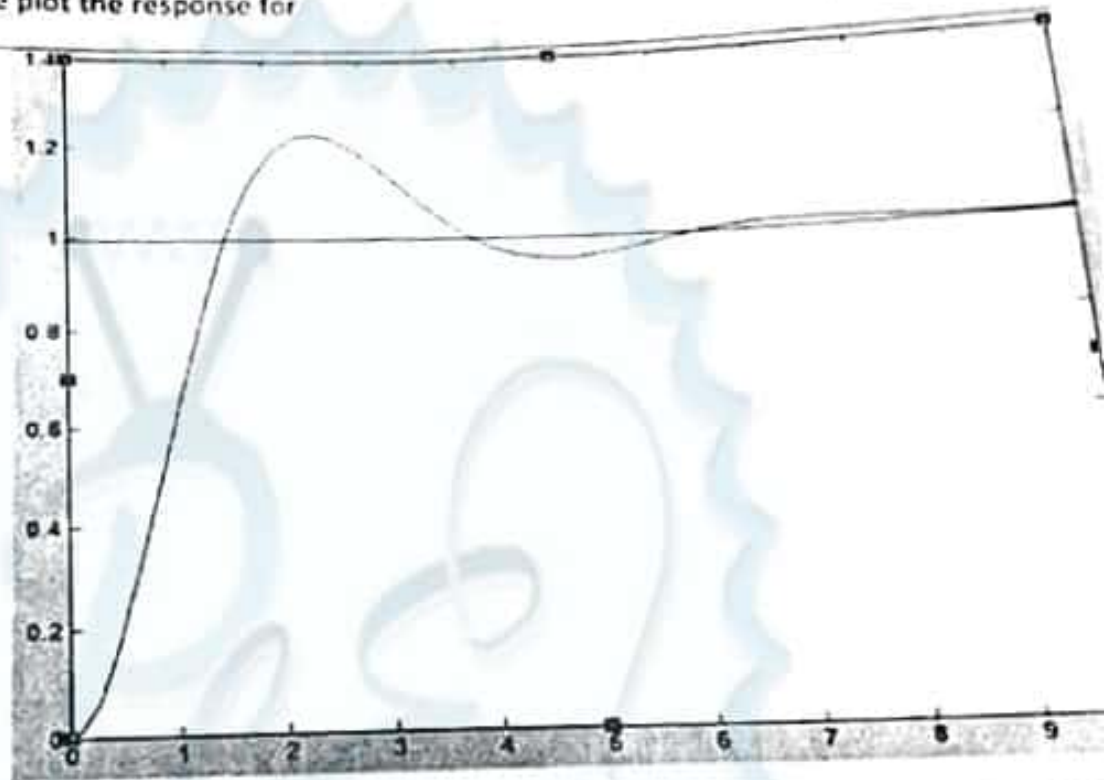
- On the same figure plot the response for

1) $r(t) = 1, \omega_n = 1, \zeta = 1$

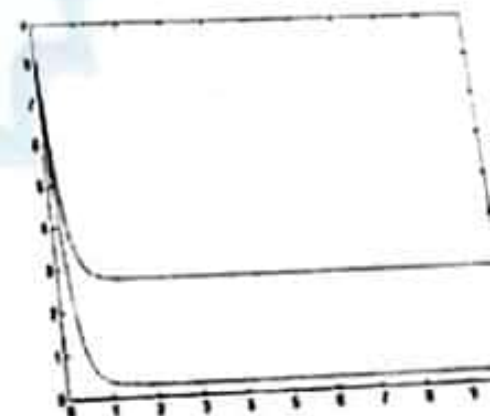
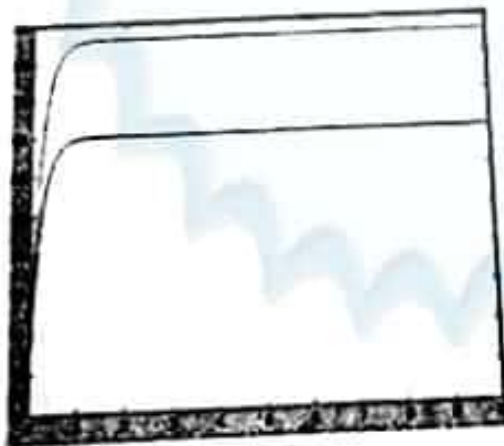
2) $r(t) = 1, \omega_n = 1, \zeta = 0$

3) $r(t) = 2, \omega_n = 1, \zeta = 0.5$

4) $r(t) = 1, \omega_n = 2, \zeta = 0.5$

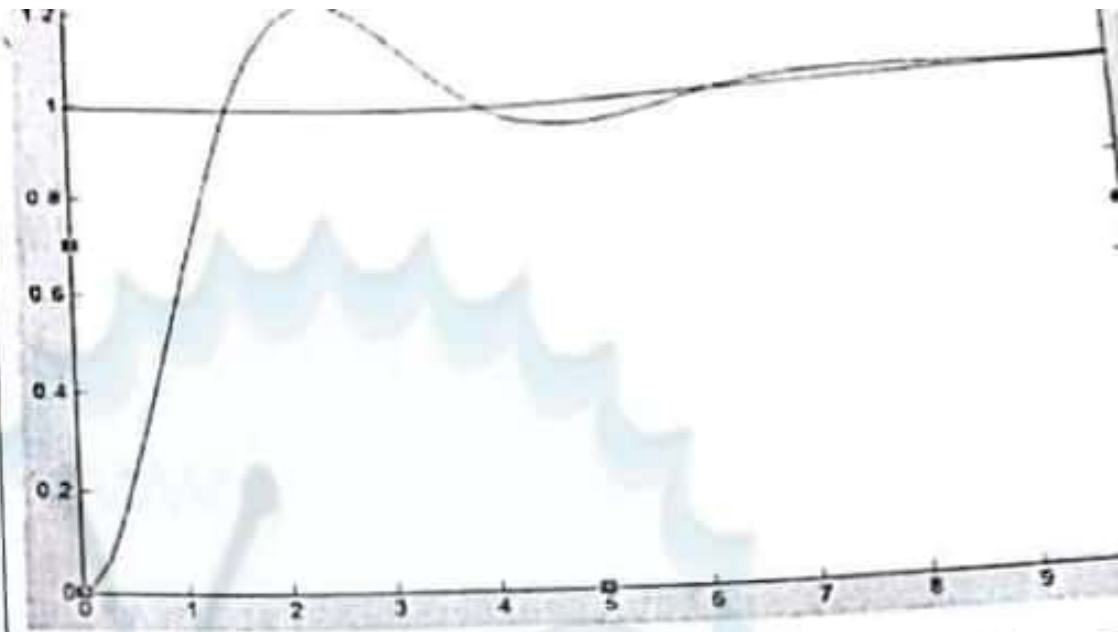


Q3) [10 marks] The following curves are the output of dc motor (open loop)

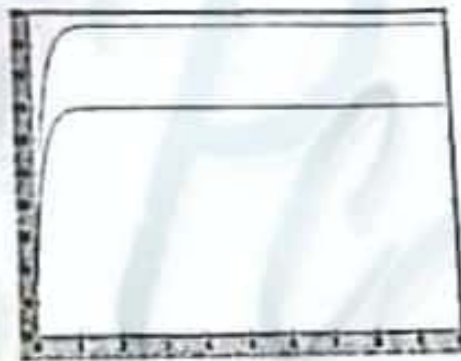


Current vs Time

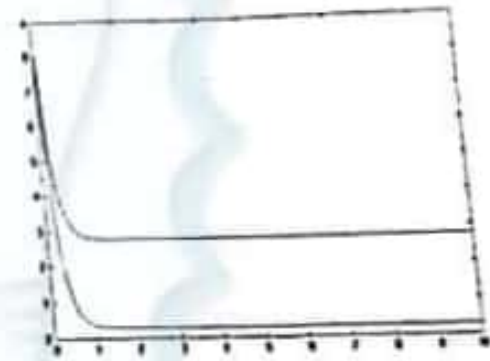
- 2) $r(t) = 1, \omega_n = 1, \zeta = 0$
 3) $r(t) = 2, \omega_n = 1, \zeta = 0.5$
 4) $r(t) = 1, \omega_n = 2, \zeta = 0.5$



Q3) [10 marks] The following curves are the output of dc motor (open loop)



Speed vs Time



Current vs Time

- Which plot represents the case of no load in each curve
- Explain the effect of load on i_a and ω
- Sketch i_a and ω in case of position closed loop system for the above case